

Nmap Network Scanning Technical Guide

TCP SYN (-SS) • UDP (-SU) • SERVICE DETECTION (-SV) • OS DETECTION (-O)

1. TCP SYN STEALTH SCAN (-SS)

TCP SCAN Default & Stealthy Reconnaissance

The TCP SYN scan is the default and most popular scan mode in Nmap. It performs a "half-open" scan by sending a **SYN** packet and listening for a response without completing the full 3-way TCP handshake (SYN-ACK / RST).

- **Mechanics:** Target responds with **SYN-ACK** = Port Open; **RST** = Port Closed; No response / ICMP unreachable = Filtered.
- **Privileges:** Requires **root** / **sudo** privileges to raw-craft IP packets.
- **Advantage:** Fast, discreet, and less likely to be logged by basic application-level loggers.

```
sudo nmap -sS 192.168.1.1 # Basic SYN scan on default top 1,000 ports
sudo nmap -sS -p 21,22,80,443,3306 192.168.1.1 # SYN scan targeted ports
```

2. UDP SCAN (-SU)

UDP SCAN Enumerating Connectionless Services

UDP scanning targets non-TCP services such as DNS (53), DHCP (67/68), SNMP (161/162), and NTP (123). Since UDP is connectionless, probing is significantly slower than TCP.

- **Mechanics:** An empty UDP packet is sent to a target port. An **ICMP Port Unreachable (Type 3, Code 3)** response means Closed. No response or service-specific UDP response means Open|Filtered.
- **Challenge:** OS rate-limiting of ICMP error messages makes full UDP scans extremely slow.
- **Best Practice:** Combine with **--top-ports** or specific port targets.

```
sudo nmap -sU --top-ports 20 192.168.1.1 # Scan top 20 most common UDP ports
sudo nmap -sU -p 53,67,123,161 192.168.1.1 # Scan specific UDP services
```

3. SERVICE & VERSION DETECTION (-SV)

VERSION DETECTION Banner Grabbing & Service Probe

Instead of simply identifying open ports based on standard numbers (e.g. port 80 = HTTP), **-sv** interrogates open ports using specific probes from **nmap-service-probes** to extract software names, exact version numbers, and application details.

- **Intensity Flags:** **--version-intensity 0-9** (Default is 7). Lower is faster; higher is more thorough.
- **Use Case:** Vital for vulnerability mapping (CVE matching against exact product versions).

```
nmap -sV 192.168.1.1 # Service version detection on open ports
nmap -sV --version-intensity 9 192.168.1.1 # Maximum intensity banner probing
```

4. OPERATING SYSTEM DETECTION (-O)

OS FINGERPRINT TCP/IP Stack Fingerprinting

Nmap sends a series of TCP and UDP packets to remote ports and analyzes subtle differences in how the remote operating system's TCP/IP stack responds (TCP window size, TCP options order, IP ID sequence generation).

- **Requirement:** Needs at least **1 open and 1 closed TCP port** on the target for high-confidence fingerprint matching.
- **Output:** Returns OS Family, Kernel Version, CPE Identifier, and OS Accuracy Percentage.

```
sudo nmap -O 192.168.1.1 # Perform remote OS fingerprinting
sudo nmap -O --osscan-guess 192.168.1.1 # Force aggressive OS guessing
```

5. CONSOLIDATED & UNIFIED COMMAND REFERENCE

SCAN OBJECTIVE	NMAP COMMAND COMBINATION	KEY CHARACTERISTICS
Full Comprehensive Scan	<code>sudo nmap -sS -sU -sV -O -F 10.0.2.3</code>	Combines SYN, Fast UDP, Versioning, and OS Detection.
Targeted Port Assessment	<code>sudo nmap -sS -sV -O -p T:21,22,80,443,3306,U:53,161 10.0.2.3</code>	Scans specific TCP/UDP ports with service banners and OS check.
Aggressive Discovery (-A)	<code>sudo nmap -A 10.0.2.3</code>	Enables OS Detection (-O), Version Detection (-sV), Script Scan (-sC), and Traceroute.
Save Scan Results (-oA)	<code>sudo nmap -sS -sV -O -oA scan_output 10.0.2.3</code>	Saves output in Normal (.nmap), XML (.xml), and Grepable (.gnmap) formats.